20概统期中（回忆版）

Part One – Select one from the given four options.

1. Assume A and B are disjoint events, $P(A)>0$, $P(B)>0$, which conclusion is correct?

(A) $P(B|A)>0$ (B) $P(A|B)=P(A)$

(C) $P(A|B)=0$ (D) $P(AB)=P(A)P(B)$

2. Assume $F(x,y)$ is the distribution function of two dimensional r.v. (X,Y), which conclusion is wrong?

(A) $F(+\infty,+\infty)=1$ (B) $F(-\infty,-\infty)=0$

(C) $F(+\infty,y)=1$ (D) $F(x,-\infty)=0$.

3. A component is manufactured through two stages. The rejection rate (the probability of defective products) is p_1 in the first stage, and p_2 in the second stage. Which option ( ) is the finished rate (the probability of qualified products)?

(A) $1-p_1-p_2$; (B) $1-p_1p_2$

(C) $1-p_1-p_2+p_1p_2$ (D) $(1-p_1)+(1-p_2)$.

4. Assume random variables X_1, X_2 have their distribution functions $F_1(x), F_2(x)$ respectively. In order to make $F(x)=aF_1(x)-bF_2(x)$ be a distribution function of some random variable, which option ( ) can make it happen?

(A) $a=\frac{3}{5}, b=-\frac{2}{5}$ (B) $a=\frac{2}{3}, b=\frac{2}{3}$

(C) $a=-\frac{1}{2}, b=\frac{3}{2}$ (D) $a=\frac{1}{2}, b=\frac{3}{2}$.

5. Assume $X\sim N(0,1)$, $Y\sim N(1,1)$, and X and Y are independent to each other, which option ( ) is correct?

(A) $P(X+Y\leq 0)=\frac{1}{2}$ (B) $P(X+Y\leq 1)=\frac{1}{2}$

(C) $P(X-Y\leq 0)=\frac{1}{2}$ (D) $P(X-Y\leq 1)=\frac{1}{2}$

Part Two – Fill in the boxes for each Question

1. If the probability for exactly one of the events A or B to happen is 0.3 (i.e., $P(A\bar{B})+P(\bar{A}B)=0.3$), and $P(A)+P(B)=0.5$, then the probability for at least one of A and B not to happen is _____.

2. Suppose A, B, C are pairwise independent events while the three events cannot happen at the same time. If each of the events happens with the same probability, then the maximum of $P(A\cup B\cup C)$ is _____.

3. A coin is tossed five times, then the ~~prob~~ probability of getting at least two heads and at least two tails is _____.

4. Suppose the random variables X and Y are independent and they both follow the exponential distribution $EXP(\lambda)$, $f_X(x)=\begin{cases}\lambda e^{-\lambda x}, & x>0\\ 0, & x\leq 0\end{cases}$. then the distribution of $\min(X,Y)$ is _____. (Please write down the distribution and its parameter.)

5. The joint probability mass function (PMF) of the random variable (X,Y) is listed as follows:

X \ Y	0	1
0	1/4	1/4
1	0	1/2

Let $F(x,y)$ be the joint CDF of (X,Y), then $F(\frac{1}{2},1)=$ _____.

6. Let $X\sim N(\mu,\sigma_1^2)$, $Y\sim N(2\mu,\sigma_2^2)$, X and Y are independent. If $P(X-Y\geq 1)=\frac{1}{2}$, then $\mu=$ _____.

7. The probability density function (PDF) of a random variable X is $f(x)=\begin{cases}e^{-x}, & x>0\\ 0, & x\leq 0\end{cases}$, then $P(X\leq 2\mid X\geq 1)=$ _____.

8. Suppose a region D is formed by the lines $x=2$, $y=2$ and the x-axis and y-axis of the coordinate system. A two-dimensional random variable (X,Y) follows a uniform distribution in the region D. Then the value of the marginal probability density function (PDF) of X at $x=1$ is _____.

9. Suppose X and Y follow Binomial Distribution $X\sim b(n,p)$, $Y\sim b(m,p)$ and be independent, then the distribution of $X+Y$ is: $X+Y\sim$ _____.

10. Suppose the random variable X follow Uniform distribution $X\sim U(a,b)$ $(a>0)$, and $P(0<X<3)=\frac{1}{4}$, $P(X>4)=\frac{1}{2}$, then $P(1<X<5)=$ _____.

Part Three – Questions and Answers

1. A shooting team has 20 shooters, of whom 4 are in the first level, 8 are in the second level, 7 are in the third level, and I is in the fourth level. The probability of each level of the ~~shoote~~ shooters entering the competition through selection is 0.9, 0.7, 0.5, 0.2. Compute the probability that a randomly selected shooter could enter the competition.

2. A hunter shoots at the first time in 100m from the prey, and the probability of hitting the prey is 0.5. If the first shooting misses, the hunter continues to shoot at the second time, and now the prey is in 150m away from the hunter. If the second shooting still misses, the hunter continues to shoot at the third time, and right now the prey is in 200m away from the hunter. If the third shooting has

not hit, the prey escapes. If the probability of the hunter hitting the prey is inversely proportional to the distance $P(X=x)=\frac{k}{x}$ (x is the distance and k is a constant), find the probability of the prey having been hit.

3. Suppose a random variable X has the density function:

$$f(x)=\begin{cases} kx(1-x), & 0<x<1, \\ 0, & \text{other} \end{cases}$$

(1) Compute the constant k;

(2) Find the value range and the density function of $Y=-3X+3$.

4. Suppose random variables X, Y follow Poison distribution $X\sim P(\lambda_1)$, $Y\sim P(\lambda_2)$. Furthermore X and Y are independent.

(1) Find the joint frequency function of X and Y;

(2) Find the conditional probability $P(X=k|X+Y=n)$, where $n\geq k$ is a non-negative integer.

[Hint: $P(X+Y=n)=\sum_{k=0}^{n} P(X=k, Y=n-k)$]

5. Suppose the two-dimensional random variable (X,Y) has the joint density function

$$f(x,y)=\begin{cases} 2(x+y), & 0<x<1, 0<y<x \\ 0, & \text{other} \end{cases},$$

(1) Find the marginal density function $F_Y(y)$;

(2) Justify the independency of X and Y, and give the explanation.

6. Suppose the random variable $X\sim U(0,1)$. Given $X=x$, the random variable Y has the conditional density function

$$f_{Y|X}(y|x)=\begin{cases} x, & 0<y<x, \\ 0, & \text{other} \end{cases}$$

(1) Find the joint density function $f(x,y)$;

(2) Find the marginal density function $f_Y(y)$;

(3) Compute $P\{X\leq Y\}$.